

Saving MQTT Protocol Documentation_251203

1、 clientid: The electricity meter uses its own meter number as the client ID.

2、 Topic

Topic can be set. The default setting is /v1/device/current meter number

(1) Scheduled data reporting topic: The currently set topic

(2) Command distribution (meter subscription, platform distribution) topic: Current set topic/cmd

(3) Command response topic: Previously configured topic/response

3、 Scheduled data reporting

(1) Topic: Current topic settings

Topic example: /v1/device/111111111111

(2) Report fields

KEY	Correlation variable description	label	unit	Data example
price				
0	No associated variables			
1	A phase voltage	"Ua"	V	219.9
2	B phase voltage	"Ub"	V	219.9
3	C phase voltage	"Uc"	V	219.9
4	A phase current	"Ia"	A	5.001
5	B phase current	"Ib"	A	5.001
6	C phase current	"Ic"	A	5.001
7	active power	"Pt"	kW	3.2997
8	Active power of Phase A	"Pa"	kW	1.0999
9	Active power in Phase B	"Pb"	kW	1.0999
10	Active power in C phase	"Pc"	kW	1.0999
11	total reactive power	"Qt"	kVar	3.2997
12	Phase A reactive power	"Qa"	kVar	1.0999
13	Phase B reactive power	"Qb"	kVar	1.0999

14	C-phase reactive power	"Qc"	kVar	1.0999
15	total apparent power	"St"	kVA	3.2997
16	A-phase power	"Sa"	kVA	1.0999
17	Phase-Viewed Power in Phase B	"Sb"	kVA	1.0999
18	C-phase power	"Sc"	kVA	1.0999
19	Total power factor	"The"		1
20	Phase A power factor	"TheA"		1
21	Phase B power factor	"TheB"		1
22	Phase C power factor	"TheC"		1
23	frequency	"Hz"	Hz	49.99
24	Positive active power quantity	kwhImp	kWh	0.01
25	Forward active power rate 1	kwhImpTariff1	kWh	0.01
26	Forward active power rate 2	kwhImpTariff2	kWh	0
27	Forward active power rate 3	kwhImpTariff3	kWh	0
28	Forward active power rate 4	kwhImpTariff4	kWh	0
29	reverse active power quantity	kwhExp	kWh	0.01
30	Reverse active power rate 1	kWhExpTariff1	kWh	0.01
31	Reverse active power rate 2	kwhExpTariff2	kWh	0
32	Reverse active power rate 3	kwhExpTariff3	kWh	0
33	Reverse active power rate 4	kwhExpTariff4	kWh	0
34	Date and Time	Time		2021-08-20 20:30:16
35	Meter number	meteraddress		1111111111
36	Message type	DataType		

37	peak point current	PeakIa	A	The maximum current value from the previous minute, updated every minute
38	continuous current	Continuou sla	A	The average current value over the past 5 minutes is updated every 5 minutes.
39	peak point voltage	PeakUa	V	The maximum voltage value over the past minute, updated every minute

(3) Example of scheduled data reporting:

```
{ "DataType": "Variable", "meteraddress": "11111111111", "Time": "2021-08-20T20:30:16", "datas": { "Ua": 647.4, "Ub": 0, "Uc": 0, "Ia": 0, "Ib": 0, "Ic": 0, "Pt": 0, "Pa": 0, "Pb": 0, "Pc": 0, "The": 0.02, "TheA": 0.02, "TheB": 1, "TheC": 1, "Hz": 49.99, "kWhImp": 0, "kWhImpTariff1": 0, "kWhImpTariff2": 0, "kWhImpTariff3": 0, "kWhImpTariff4": 0, "kWhExp": 0, "kWhExpTariff1": 0, "kWhExpTariff2": 0, "kWhExpTariff3": 0, "kWhExpTariff4": 0 } }
```

4、 Command distribution (meter subscription, platform distribution)

5、 (1) topic: the current set topic cmd

Topic example: /v1/device/11111111111/cmd

(2) Command field for sending

KEY	Correlation variable description	label	form	Data example
price				
0	No associated variables			
1	Meter number	"meteraddress"	XXXXXXXXXXXX	11111111111
2	order ID	"messageID"		
3	order	"cmd"		"relaycontrol" "getrelaystatus"

				"readvariable" "setip" "getip" "setfreq" "getfreq" "setmqttpara" "getmqttpara"
4	Execution result	"cmdresult"		0: Success 1: Failed
5	Relay status	"relaystatus"		"on" : switch on "Off": Circuit breaker tripped
6	Time Zone Table	" ZONETABLE"	MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number)	Supports up to 14 time zones

			MM (Month) DD (Day) NN (Schedule Number) MM (Month) DD (Day) NN (Schedule Number)	
7	Schedule Table	" SCHEDUL ETABLE"	hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (rate number) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code) hh (hours) mm (minutes) NN (tariff code)	Supports up to 14 time slots

(3) Get relay status:

Issue command:

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get  
relaystatus"}
```

reply :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmdresult  
":0," cmd" : " getrelaystatus" ," relaystatus" : " on" }
```

(4) closing command

The meter closes when the message content is relayon, and opens when the message content is relayoff.

Power cut-off command example:

```
{"meteraddress":"210924010003","messageID":"1234560","cmd":"relaycontrol","message":"relayoff"}
```

Example of closing command:

```
{"meteraddress":"210924010003","messageID":"1234560","cmd":"relaycontrol","message":"relayon"}
```

(5) Read real-time data

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"readvariable"}
```

reply :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmdresult":0,"cmd":"readvariable",  
"datas":{"Ua":647.4,"Ub":0,"Uc":0,"Ia":0,"Ib":0,"Ic":0,"Pt":0,"Pa":0,"Pb":0,"Pc":0,"The":0.02,"TheA":0.02,"TheB":1,"TheC":1,"Hz":49.99,"kWhImp":0,"kWhImpTariff1":0,"kWhImpTariff2":0,"kWhImpTariff3":0,"kWhImpTariff4":0,"kWhExp":0,"kWhExpTariff1":0,"kWhExpTariff2":0,"kWhExpTariff3":0,"kWhExpTariff4":0}}
```

(6) Configure IP port

order :

```
{"meteraddress":"165125062053","messageID":"2022121013091201","cmd":"setip","HOST":"117.84.15.151","PORT":18888}
```

reply

```
meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":0,"cmd":"setip"}
```

(7) Copy IP port

order :

```
{"meteraddress":"165125062053","messageID":"2022121013091201","cmd":"getip"}
```

reply :

```
{"meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":0,"cmd":"getip","HOST":"39.104.65.88","PORT":1883}
```

(8) Set reporting cycle

order :

```
{"meteraddress":"165125062053","messageID":"2022121013091201","cmd":"set  
freq","FREQ":900}
```

reply :

```
{"meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":  
"0","cmd":"setfreq"}
```

(9) Read and report cycle

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get  
freq"}
```

reply :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmdresult":  
"0," cmd" : " getfreq" ," FREQ" :300}
```

(10) Set the MQTT connection username and password

order :

```
{"meteraddress":"102305290001","messageID":"2022121013091201","cmd":"set  
mqttpara","USERNAME":"sav","PASSWORD":"XUMQTT","TOPIC":"/v1/device/1  
02305290001"}
```

Use the response topic (original topic/response)

reply :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmdresult":  
"0," cmd" : " setmqttpara" }
```

(11) Copy the MQTT connection username and password

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get  
mqttpara"}
```

reply :

```
{"meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":  
"0","cmd":"getmqttpara","USERNAME":"sav","PASSWORD":"XUMQTT"}
```

(12) set-up time

order :

```
{"meteraddress":"165125062053","messageID":"2022121013091201","cmd":"setdatetime","DATETIME":"2025-09-02T15:00:00"}
```

reply :

```
{"meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":0,"cmd":"setdatetime" }
```

(13) Reading time

order :

```
{"meteraddress":"165125062053","messageID":"2022121013091201","cmd":"getdatetime"}
```

reply :

```
{"meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":"0","cmd":"getdatetime","DATETIME":"2025-09-02T15:13:00"}
```

(14) Switch time when reading the time zone table

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"getzoneswitchtime"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult":"0","cmd":"getzoneswitchtime","ZONESWITCHTIME":"0000000000"}
```

(15) Set time zone table switch time

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"setzoneswitchtime","ZONESWITCHTIME":"2601010000"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult":"0","cmd":"setzoneswitchtime"}
```

(16) Copy schedule time switch

order :


```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get  
scheduleswitchtime"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult  
":"0","cmd":"getscheduleswitchtime","SCHEDULESWITCHTIME":"0000000000"}
```

(17) Set schedule table switching time

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"setsc  
heduleswitchtime","SCHEDULESWITCHTIME":"2601010000"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult":  
"0","cmd":"setscheduleswitchtime"}
```

(18) Number of time zones to read

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"getz  
onenumbers"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult":  
"0","cmd":"getzonenumbers","ZONENUMBER":"1"}
```

(19) Set number of time zones

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"set  
zonenumbers","ZONENUMBER":"2"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult  
":"0","cmd":"setzonenumbers"}
```

(20) Number of reading date schedules

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get  
schedulesubnumber"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult":
":0","cmd":"getschedulablenumber","SCHEDULETABLENUMBER":"1"}
```

(21) Set number of time slots

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"sets
chedulablenumber","SCHEDULETABLENUMBER":"8"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult
":0,"cmd":"setschedulablenumber"}
```

(22) Number of reading days

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get
schedulingnumber"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult
":0,"cmd":"getschedulingnumber","SCHEDULENUMBER":"5"}
```

(23) Set time period count

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"sets
chedulingnumber","SCHEDULENUMBER":"8"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult
":0,"cmd":"setschedulingnumber"}
```

(24) Read the first set of time zone tables

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get
zonetable1"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult
":0,"cmd":"getzonetable1","ZONETABLE":"010101010101"}
```

(25) Set the first time zone table

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"set  
zonetable1","ZONETABLE":"010101060102"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult  
":"0","cmd":"setzonetable1"}
```

(26) Read the second set of time zone tables

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get  
zonetable2"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult  
":"0","cmd":"getzonetable2","ZONETABLE":"010101010101"}
```

(27) Set up the second time zone table

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"set  
zonetable2","ZONETABLE":"010101090103"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult  
":"0","cmd":"setzonetable2"}
```

(28) Read the first set of schedule for Day 1

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get  
scheduletable1_1"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult  
":"0","cmd":"getscheduletable1_1","SCHEDULETABLE":"00000408000212000317  
0002210003210003210003210003"}
```

(29) Set the first set of schedule for the first day

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"sets
```

```
cheduletable1_1","SCHEDULETABLE":"050003080002110004130002150003170002210003230004"}}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult":"0","cmd":"setscheduletable1_1"}
```

(30) Read the first set of schedules for the second day

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"getscheduletable1_2"}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult":"0","cmd":"getscheduletable1_2","SCHEDULETABLE":"000004080002120003170002210003210003210003210003"}
```

(31) Set the second-day schedule for the first set

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"setscheduletable1_2","SCHEDULETABLE":"050003080002110004130002150003170002210003230004"}}
```

reply :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult":"0","cmd":"setscheduletable1_2"}
```

(32) To retrieve the schedule tables for the first set: -Day 3: cmd getscheduletable1_3-Day 4: cmd getscheduletable1_4-Day 5: cmd getscheduletable1_5-Day 6: cmd getscheduletable1_6-Day 7: cmd getscheduletable1_7-Day 8: cmd getscheduletable1_8

(33) Set the first set's 3rd-day schedule command as setscheduletable1_3, the 4th-day schedule command as setscheduletable1_4, the 5th-day schedule command as setscheduletable1_5, the 6th-day schedule command as setscheduletable1_6, the 7th-day schedule command as setscheduletable1_7, and the 8th-day schedule command as setscheduletable1_8.

(34) Read the second set of schedule for Day 1

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get
```

scheduletable2_1"}
}

respond :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult":
":0","cmd":"getscheduletable2_1","SCHEDULETABLE":"01010109010309010309
0103090103090103090103090103"}
```

(35) Set the first day schedule for the second set

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"sets
cheduletable2_1","SCHEDULETABLE":"0500030800021100041300021500031
70002210003230004"}
```

respond :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult
":0","cmd":"setscheduletable2_1"}
```

(36) Read the second set of schedules for the second day

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"get
scheduletable2_2"}
```

respond :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult
":0","cmd":"getscheduletable2_2","SCHEDULETABLE":"00000408000212000
3170002210003210003210003210003"}
```

(37) Set the second set of schedules for the second day

order :

```
{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"sets
cheduletable2_2","SCHEDULETABLE":"05000308000211000413000215000317000
2210003230004"}
```

respond :

```
{"messageID":"2022121013091201","meteraddress":"111111111111","cmdresult
":0","cmd":"setscheduletable2_2"}
```

(38) To retrieve the schedule tables for the second set: -Day 3: cmd getscheduletable2_3-Day 4: cmd getscheduletable2_4-Day 5: cmd getscheduletable2_5-Day 6: cmd getscheduletable2_6-Day 7: cmd getscheduletable2_7-Day 8: cmd getscheduletable2_8

(39) Set the third-day schedule command for the second set as `setscheduletable2_3`, the fourth-day schedule command as `setscheduletable2_4`, the fifth-day schedule command as `setscheduletable2_5`, the sixth-day schedule command as `setscheduletable2_6`, the seventh-day schedule command as `setscheduletable2_7`, and the eighth-day schedule command as `setscheduletable2_8`.